

Your grade: 87.50%

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Next item →

1. Consider the digits, 1, 1, 1, 2, 2, 2, 3. How many 6 digit numbers exists using the following digits such that it starts with 1 or it ends with 2?

1 / 1 point

- ☐ 36
- ☒ 38
- ☐ 32
- ☐ 34

👍 Nice work

Assume A is the set of numbers that starts with 1. $|A| = \frac{2 \cdot 4 \cdot 3 \cdot 2 \cdot 3 \cdot 1}{1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 1} = 20$

Also assume B is the set of numbers that ends with 2. $|B| = \frac{1 \cdot 4 \cdot 3 \cdot 2 \cdot 2 \cdot 3}{2 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 1} = 30$.

$$|A \cap B| = \frac{2 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot 3}{1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 1} = 12$$

$$|A \cup B| = 20 + 30 - 12 = 38.$$

2. Assume we want to split 6 different books to 3 people such that each person receives at least 1 book.

1 / 1 point

- ☒ 540
- ☐ 28
- ☐ 534
- ☐ 10

👍 Nice work

We want to count the number of onto functions from a domain of 6 elements to a range of 3 elements. This can be calculated as $3^6 - 3 \cdot 2^6 + 3 = 540$

3. Consider two sets $A = \{a_1, a_2, a_3, a_4\}$ $B = \{b_1, b_2, b_3, b_4\}$. How many functions exists from A to B such that it's not onto and not a bijection.

1 / 1 point

- ☐ 24
- ☒ 232
- ☐ 40
- ☐ 104

👍 Nice work

As A and B are both having 4 elements, any bijection is also onto. We know that the number of bijections from A to B is $4!=24$. So, the number of functions that are not onto and not bijection is equal to $4^4 - 24 = 232$.

4. At a conference with 120 people, everyone speaks at least one of English, Spanish, or French.

0 / 1 point

75 speak English, 60 speak Spanish, and 45 speak French.

What is the maximum number of people who speak exactly one language?

60

👎 Try again

5. Consider the set $A = \{1, 2, 3, \dots, 600\}$. How many elements exists in this set such that it is not a multiple of 2 and also not a multiple of 3?

1 / 1 point

- ☒ 200
- ☐ 240
- ☐ 400
- ☐ 360

👍 Nice work

Assume x is the number of elements divisible by 2 which is $x = \frac{600}{2} = 300$ and assume y is the number of elements divisible by 3 which is $y = \frac{600}{3} = 200$.

We also know that $|x \cap y|$ is the set of elements that are multiple of both 2 and 3 (in other words it's a multiple of 6 which is $\frac{600}{6} = 100$.

We know that $|x \cup y| = |x| + |y| - |x \cap y| = 300 + 200 - 100 = 400$. Now this is the number of numbers that are either divisible by 2 or by 3. So we need to calculate $|S| - |x \cup y| = 600 - 400 = 200$.

6. Consider the set $A = \{1, 2, 3, \dots, 120\}$. How many elements exist in this set such that it is a multiple of 2 or a multiple of 3 or a multiple of 5?

1 / 1 point

- ☐ 92
- ☒ 88
- ☐ 90
- ☐ 86

👍 Nice work

Assume x is the number of elements divisible by 2 which is $x = \frac{120}{2} = 60$ and assume y is the number of elements divisible by 3 which is $y = \frac{120}{3} = 40$ and assume z is the number of elements divisible by 5 which is $z = \frac{120}{5} = 24$.

Similar to previous questions, we can easily see that:

$$|x \cap y| = \frac{120}{6} = 20$$

$$|x \cap z| = \frac{120}{10} = 12$$

$$|y \cap z| = \frac{120}{15} = 8$$

$$|x \cap y \cap z| = \frac{120}{30} = 4$$

$$\text{We know that } |x \cup y \cup z| = 60 + 40 + 24 - 20 - 12 - 8 + 4 = 88.$$

7. Consider the set $A = \{1, 2, 3, \dots, 120\}$. How many elements exists in this set such that it is a multiple of 2 or a multiple of 3?

1 / 1 point

- ☒ 80
- ☐ 90
- ☐ 60
- ☐ 70

👍 Nice work

Assume x is the number of elements divisible by 2 which is $x = \frac{120}{2} = 60$ and assume y is the number of elements divisible by 3 which is $y = \frac{120}{3} = 40$.

We also know that $|x \cap y|$ is the set of elements that are multiple of both 2 and 3 (in other words it's a multiple of 6 which is $\frac{120}{6} = 20$.

$$\text{We know that } |x \cup y| = |x| + |y| - |x \cap y| = 60 + 40 - 20 = 80.$$

8. Assume 4 people are standing in a line. In how many ways they can stand in a different way such that none of them are standing in the same position as their previous standing.

1 / 1 point

- ☐ 12
- ☐ 11
- ☐ 10
- ☒ 9

👍 Nice work

We must count the number of derangements using inclusion-exclusion which will be equal to 9